

Discussion: Chain Rule and Product Rule



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chain rule

$$\frac{d}{dx} f[g(x)] = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

outer derivative inner derivative

derivatives dx , dy were derived from finite differences Δx and Δy

$$\frac{df}{dx} = \frac{df}{dg} \frac{dg}{dx}$$

outer derivative inner derivative

$$\frac{df}{dx} = \frac{\frac{df}{dg}}{\frac{dx}{dg}}$$

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chain rule

$$\frac{d}{dx} f[g(x)] = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

outer derivative inner derivative

example:

$$f(t) = e^{x(t)} \quad x(t) = \cos(\omega t + \varphi)$$

$$\frac{df}{dt} = \frac{df}{dx} \frac{dx}{dt} = e^{x(t)} [-\sin(\omega t + \varphi)] \omega$$

product rule

$$\frac{d}{dx} [f(x)g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$$

$$f(x)g(x) = \int g(x) \frac{d}{dx} f(x) dx + \int f(x) \frac{d}{dx} g(x) dx \quad fg = \int gf' + \int fg'$$

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product rule

$$\frac{d}{dx}[f(x)g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x)$$

$$fg = \int gf' + \int fg'$$

$$f(x)g(x) = \int g(x)\frac{d}{dx}f(x)dx + \int f(x)\frac{d}{dx}g(x)dx$$

$$\int gf' = fg - \int fg'$$

$$\int_{\tau=0}^{\tau=\infty} \tau \vartheta e^{-\vartheta \tau} d\tau = \int_{\tau=0}^{\tau=\infty} \tau \vartheta e^{-\vartheta \tau} d\tau = \vartheta \left[\int_{\tau=0}^{\tau=\infty} \tau e^{-\vartheta \tau} d\tau \right] = \vartheta \left[\tau \frac{e^{-\vartheta \tau}}{-\vartheta} \Big|_{\tau=0}^{\tau=\infty} - \int_{\tau=0}^{\tau=\infty} \frac{e^{-\vartheta \tau}}{-\vartheta} d\tau \right]$$

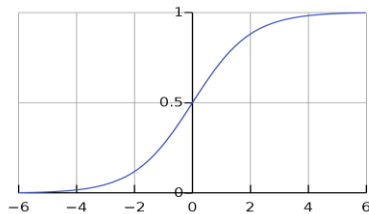
$$= \vartheta \left[\tau \frac{e^{-\vartheta \tau}}{-\vartheta} \Big|_{\tau=0}^{\tau=\infty} - \frac{e^{-\vartheta \tau}}{\vartheta^2} \Big|_{\tau=0}^{\tau=\infty} \right]$$

$$= \vartheta \left[\infty \frac{e^{-\vartheta \infty}}{-\vartheta} - 0 \frac{e^{-\vartheta 0}}{-\vartheta} - \frac{e^{-\vartheta \infty}}{\vartheta^2} + \frac{e^{-\vartheta 0}}{\vartheta^2} \right]$$

$$= \vartheta \left[0 - 0 + \frac{1}{\vartheta^2} \right] = \frac{1}{\vartheta}$$

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sigmoid:



$$(-\infty; +\infty) \rightarrow (0; 1)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{1 + e^x}$$

$$= \frac{e^x + 1 - 1}{1 + e^x}$$

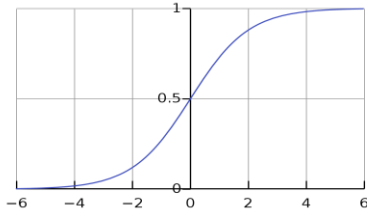
$$= \frac{e^x + 1}{1 + e^x} - \frac{1}{1 + e^x}$$

$$= 1 - \frac{1}{1 + e^x} = \mathbf{1 - \sigma(-x)}$$

$$\sigma(x) = \mathbf{1 - \sigma(-x)}$$

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sigmoid:



$$(-\infty; +\infty) \rightarrow (0; 1) \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

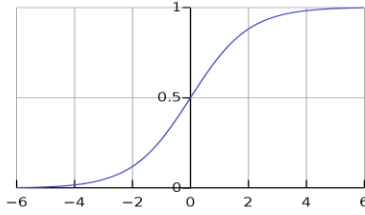
$$y := 1 + e^{-x}$$

$$\begin{aligned} \frac{d\sigma(x)}{dx} &= \frac{d\sigma(y[x])}{dx} = \frac{d\sigma}{dy} \frac{dy}{dx} = \frac{-1}{y^2} (-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2} \\ &= \frac{e^{-x}}{(1 + e^{-x})(1 + e^{-x})} = \frac{e^{-x}}{(1 + e^{-x})} \frac{1}{(1 + e^{-x})} \\ &= \frac{e^{-x} + 1 - 1}{1 + e^{-x}} \frac{1}{(1 + e^{-x})} \\ &= \left[1 - \frac{1}{1 + e^{-x}} \right] \frac{1}{(1 + e^{-x})} \\ &= [1 - \sigma(x)] \sigma(x) \end{aligned}$$

$$\frac{d\sigma(x)}{dx} = [1 - \sigma(x)] \sigma(x)$$

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sigmoid:



$$(-\infty; +\infty) \rightarrow (0; 1) \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\frac{d\sigma(x)}{dx} = [1 - \sigma(x)]\sigma(x)$$

$$\mathcal{L} = \frac{1}{2}(y - \hat{y})^2 \quad \hat{y} = \sigma(x) \quad x = \sum_i I_i w_i + b$$

$$\frac{d\mathcal{L}}{dw_i} = \frac{d\mathcal{L}}{d\hat{y}} \frac{d\hat{y}}{dx} \frac{dx}{dw_i} = -(y - \hat{y}) [1 - \sigma(x)]\sigma(x) I_i$$

$$\frac{d\mathcal{L}}{db} = \frac{d\mathcal{L}}{d\hat{y}} \frac{d\hat{y}}{dx} \frac{dx}{db} = -(y - \hat{y}) [1 - \sigma(x)]\sigma(x) 1$$

$$\frac{d\mathcal{L}}{dI_i} = \frac{d\mathcal{L}}{d\hat{y}} \frac{d\hat{y}}{dx} \frac{dx}{dI_i} = -(y - \hat{y}) [1 - \sigma(x)]\sigma(x) w_i$$

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